



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 20- 402 - 2

Date: 12-10-2021

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SPECIFICATION

FOR

Meter Enclosures

Issue: Oct -2021/ Rev- 1



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1 Meter Enclosures - General Requirements

This Enclosures Specification covers manufacturing, testing, and supply of Meter Enclosures for three-phase direct connected meters.

- a- The Meter Enclosure shall be made of insulating material intended to contain meters to be installed on walls or similar flat surface structures and include a Miniature Circuit Breaker (MCB).
- b- The Meter Enclosure shall be weather proof and ultraviolet resistant (to within type test limits, at minimum).
- c- The Meter Enclosure will be supplied with the MCB and the MCB will be mounted inside the Meter Enclosure in such a way as to permit access to the MCB lever without opening the Meter Enclosure cover.
- d- The Meter Enclosures shall be compatible with the supplied Smart Meter case dimensions for the required installation configurations.

1.1. Service Condition

1.1.1 The Meter Enclosures to be supplied according to this specification shall be suitable for satisfactory continuous operation under the following environmental conditions.

	Condition	Specification
1	Maximum ambient temperature (°C)	70
2	Minimum ambient temperature (°C)	-5
3	Maximum daily average temperature (°C)	35
4	Maximum annual average temperature (°C)	30
9	Average relative humidity (%)	80
10	Maximum wind velocity (km/h)	180

1.2. Plastic Material and Construction Requirement

- a) The base and cover of the Meter Enclosure shall be made of injection molded Polycarbonate with the cover fully transparent. The base of the Meter Enclosure shall be grey in color. The thickness shall be minimum 3 mm for three phase.
- b) The Meter Enclosure for Pole mounted transformer shall have IP 54, and inclined roof not less than 5°.
- c) The Meter Enclosure shall have a tapered roof for easy flow of rainwater.
- d) The Meter Enclosure shall have suitable holes of diameter 25mm² at the bottom / side for entry of service connection wires fixed with two rubber grommets.
- e) The Meter Enclosure must be unbreakable, and shall withstand temperatures up to 125°C, certified and warranted by the manufacturer.
- f) Clearance of minimum 10 mm shall be maintained from top and both sides of the meter.
However, the clearance at bottom will be minimum 60 mm for the phase including the MCB.



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- g) The overall dimensions (in mm) of the Meter Enclosure will be as per the dimensions mentioned above considering bidders own meter.
- h) For holding and locking the cover, U-shaped latches with one hole for reverting on the base and other for sealing the Meter Enclosure shall be provided. This clamp will also clamp the cover with base. The total number of clamps for each Meter Enclosure will be two. In addition, sealing holes of 3 to 3.5 mm diameter will be provided in U clamps.
- i) The Meter Enclosure shall have two hinges.
- j) The Meter Enclosure shall be provided with four installation/attachment holes of 8 mm dia.
- k) The meter base support shall be raised by about 10 mm in the Enclosure for easy wiring. Soft rubberized gasket for protection on all around base shall be provided.
- l) The Meter Enclosure shall meet IP 54 for dust and water protection as per IEC 60529:1989. The following information shall be clearly embossed on the Meter Enclosures
- m) Distribution System Company (DISCO) Name
- n) The Meter Enclosure shall have an enough space for placing the consumer identification card, which contains meter identification number (e.g., serial number), customer name, and connection registration number on front of the Meter Enclosure cover.

1.3. Tests and Compliance standards

1.3.1 General – The tests shall be made at the prevailing ambient temperature unless specified otherwise in the relevant clause.

1.3.2 Categories of Test- The test shall constitute type and acceptance tests as enumerated below.

No.	Standard	Description
1	IEC 60670-1(1989)	The Meter Enclosure shall be designed per IEC, i.e. the engineering plastic used shall not support combustion (Fire Retardant). Type Tests (a) Test for dimensions (b) Test for Mechanical strength. (c) Test for water absorption. (d) Test for stability at high temperature.
2	IEC 60898-1 (2002)	Test for resistance to abnormal Heat and fire. Glow wire test at 650 deg. as per Publication No. IEC 60695-2-11
3	IEC 60323	Fire retardant test and test for self-extinguishing Property.
4	IEC 60707-1981	Test for determination of flammability of solid electrical insulating materials when exposed to igniting sources. Material shall confirm to category
5	IEC 61338	Verification of di-electric properties.
6	Insulation Test With 500V Megger.	Insulation test with 500VDC Megger for one-minute insulation should be 100 M ohms.